

Q12. VLSI Circuit Partitioning using Randomized Min-Cut

Given a circuit netlist graph, partition the cells into two nearly balanced groups while minimizing the number or weight of cut edges. Compare a randomized contraction approach with a deterministic greedy partitioning method.

Let the circuit be represented by a weighted graph

$$G = (V, E)$$

where each vertex represents a module/cell and w_{ij} represents the interconnection cost between modules i and j .

Define

$$x_i = \begin{cases} 0, & \text{if module } i \text{ is assigned to partition } A \\ 1, & \text{if module } i \text{ is assigned to partition } B \end{cases}$$

The cut cost can be written as

$$\min Z = \sum_{(i,j) \in E} w_{ij} |x_i - x_j|$$

because $|x_i - x_j| = 1$ only when the two modules belong to different partitions.

For balanced partitioning,

$$L \leq \sum_{i=1}^n x_i \leq U$$

where L and U specify the allowable size of partition B .

For equal bipartition,

$$\sum_{i=1}^n x_i = \frac{n}{2}$$

for even n , with

$$x_i \in \{0, 1\}.$$

Thus,

$$\min \text{Inter-partition communication}$$

subject to balanced partition sizes.

Q13. Power-Aware VLSI Placement

Each cell has power consumption and thermal contribution. Design a randomized placement algorithm that minimizes both wire length and thermal hotspots.

Let module i be assigned location

$$P_i = (x_i, y_i)$$

and let c_{ij} denote the communication weight between modules i and j .

The total Manhattan wire length is

$$WL = \sum_{(i,j) \in E} c_{ij} (|x_i - x_j| + |y_i - y_j|).$$

Let $T(P)$ denote the thermal cost or hotspot penalty of placement P .

A multi-objective formulation is

$$\boxed{\min Z = \alpha WL + \beta T(P)}$$

where α and β represent the importance of communication and temperature.

Subject to unique placement,

$$P_i \neq P_j, i \neq j.$$

Each module must be assigned to an allowed chip location:

$$P_i \in \mathcal{L}.$$

If a maximum allowable temperature is imposed,

$$T_k(P) \leq T_{\max}, \forall k.$$

The placement must also remain inside the chip boundary:

$$0 \leq x_i \leq X_{\max}, 0 \leq y_i \leq Y_{\max}.$$

Hence the goal is

$$\boxed{\text{Minimize wire length and thermal concentration simultaneously.}}$$

Q14. Randomized Floorplanning using Simulated Annealing

Given rectangular VLSI modules with fixed dimensions, place them inside a chip region while minimizing area and interconnection length. Allow random swaps, rotations, and moves.

Suppose each module i has width w_i , height h_i , and lower-left coordinate (x_i, y_i) .

Let the enclosing chip dimensions be W_c and H_c .

A common objective is

$$\min Z = \alpha A + \beta WL$$

where

$$A = W_c H_c$$

is the total floorplan area and

$$WL = \sum_{(i,j) \in E} c_{ij} (|x_i - x_j| + |y_i - y_j|)$$

is the estimated wire length.

For any two modules i and j , they must not overlap. At least one of the following must hold:

$$x_i + w_i \leq x_j$$

or

$$x_j + w_j \leq x_i$$

or

$$y_i + h_i \leq y_j$$

or

$$y_j + h_j \leq y_i.$$

Boundary constraints are

$$0 \leq x_i, x_i + w_i \leq W_c$$

and

$$0 \leq y_i, y_i + h_i \leq H_c.$$

If rotation is allowed,

$$r_i \in \{0, 1\}$$

may represent whether a module is rotated by 90° .

A richer formulation can also include delay:

$$\min Z = \alpha A + \beta WL + \gamma D.$$

Q15. Thermal-Aware Task Mapping on Network-on-Chip

Model a Network-on-Chip as a graph and map tasks to processing elements using a randomized algorithm. Minimize communication cost and temperature imbalance.

Let there be n tasks and m processing elements (PEs).

Define

$$x_{ik} = \begin{cases} 1, & \text{if task } i \text{ is mapped to PE } k \\ 0, & \text{otherwise.} \end{cases}$$

Let c_{ij} be the communication volume between tasks i and j , and d_{kl} be the communication distance between PEs k and l .

The communication cost is

$$C = \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^m \sum_{l=1}^m c_{ij} d_{kl} x_{ik} x_{jl}.$$

Let $T_{\max}$ denote the maximum temperature among all processing elements.

Then a thermal-aware objective is

$$\min Z = \alpha C + \beta T_{\max}$$

or alternatively

$$\min Z = \alpha C + \beta \sum_{k=1}^m (T_k - \bar{T})^2$$

to reduce thermal imbalance.

Each task must be assigned exactly once:

$$\sum_{k=1}^m x_{ik} = 1, \forall i.$$

If each PE can host only one task,

$$\sum_{i=1}^n x_{ik} \leq 1, \forall k.$$

If PEs have computational capacity R_k and task i requires r_i ,

$$\sum_i r_i x_{ik} \leq R_k.$$

Thermal constraints may be imposed as

$$T_k \leq T_{\text{safe}}, \forall k.$$

Thus the optimization balances Communication Cost $\leftrightarrow$ Thermal Safety

Q16. Gate Sizing Optimization

Assume gate i can be implemented using one of K_i available sizes.

Define

$$x_{ik} = \begin{cases} 1, & \text{if gate } i \text{ uses size } k \\ 0, & \text{otherwise.} \end{cases}$$

Let

- $D(x)$ = circuit delay,
- $P(x)$ = total power,
- $A(x)$ = total area.

A weighted multi-objective formulation is

$$\min Z = \alpha D(x) + \beta P(x) + \gamma A(x)$$

with

$$\alpha + \beta + \gamma = 1.$$

Exactly one size must be selected for each gate:

$$\sum_{k=1}^{K_i} x_{ik} = 1, \forall i.$$

Binary restriction:

$$x_{ik} \in \{0, 1\}.$$

If a_{ik} denotes the area of gate i using size k ,

$$\sum_i \sum_k a_{ik} x_{ik} \leq A_{\max}.$$

Similarly, if p_{ik} is power consumption,

$$\sum_i \sum_k p_{ik} x_{ik} \leq P_{\max}.$$

For a timing-constrained design,

$$D(x) \leq D_{\max}.$$

Therefore, the general formulation is

$$\min[\alpha D + \beta P + \gamma A]$$

subject to

$$D \leq D_{\max}, P \leq P_{\max}, A \leq A_{\max}.$$

17. **FPGA Logic Block Placement:** Given configurable logic blocks and pairwise connection costs, assign them to FPGA locations. Use randomized local search to reduce communication distance.
18. **Pin Assignment Optimization:** Given a set of circuit pins and required connections, assign logical signals to physical pins so that total routing cost is minimized. Use random permutation and swap-based improvement.
19. **Randomized Bus Routing:** Route a set of related signals in parallel while minimizing total wire length, crossing, and congestion. Randomize path selection and compare with shortest-path-only routing.
20. **Randomized Test Pattern Generation:** Generate random input vectors for a combinational circuit and estimate fault-detection coverage. Compare purely random testing with biased-random testing.

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Students are required to complete **at least two programming assignments** from the given set of VLSI optimization problems. Each selected assignment must include the **problem formulation, decision variables, objective function, constraints, solution representation, randomized algorithm design, implementation, sample input/output, and performance analysis**. The programs should be implemented independently, and the obtained results should be compared across multiple runs using different random seeds wherever applicable.